A Presentation of enrollment figures at Dalhousie University Faculty of Computer Science

Comparing female undergraduate enrolment with female graduate enrolment for FCS, for the years 2021, 2023, and 2025. See Table 1 and Figure 1

Table 1: Female undergraduate vs. graduate students

Faculty of Computer Science Undergraduate and Graduate Enrollment figures by Gender		
Academic Year	Female	
	Undergraduate	**Graduate**
2021/2022	288	183
2023/2024	312	176
2025/2026	247	124

Sources:
Dalhousie University, Registrar's Office. (2021). *2021 enrollment by gender* [PDF report]. 2021 Report
Dalhousie University, Registrar's Office. (2023). *2023 enrollment by gender* [PDF report]. 2023 Report
Dalhousie University, Registrar's Office. (2025). *2025 enrolment by gender* [PDF report]. 2025 Report

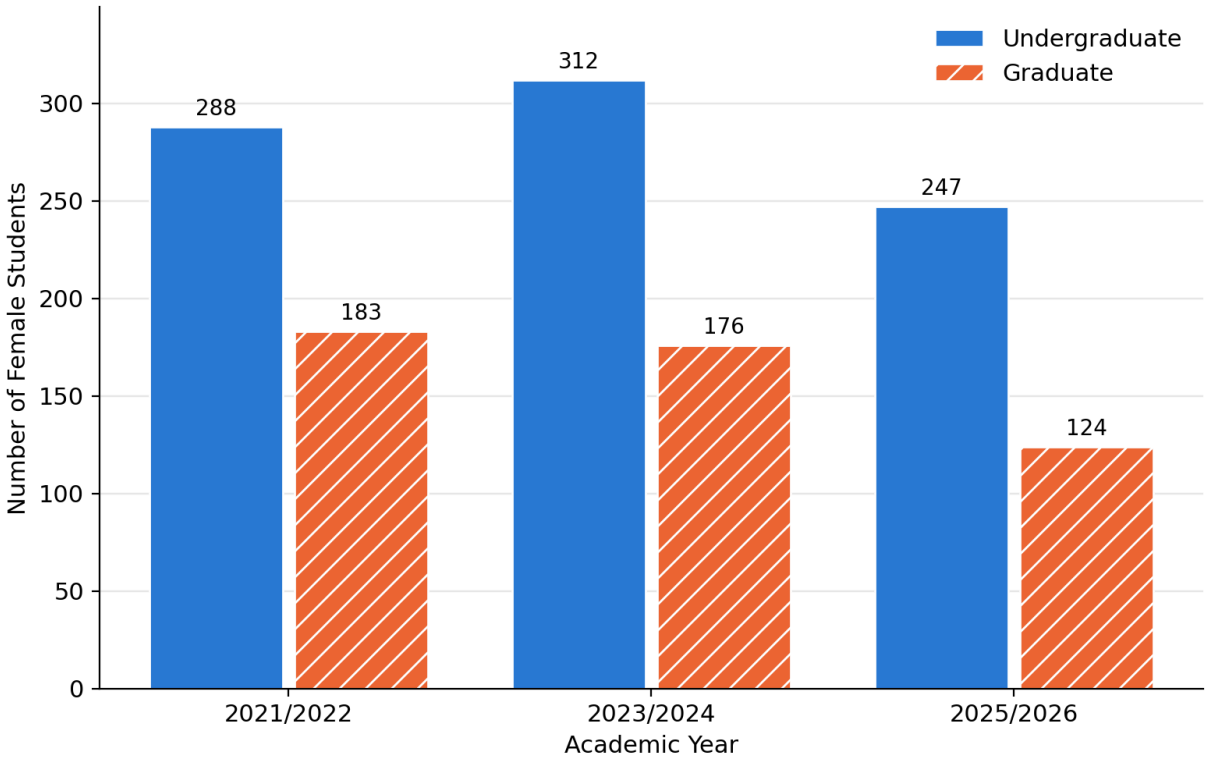


Figure 1: Female Undergraduate vs. Graduate Enrolment in FCS

Justification for Grouped Bar Chart

The task is to compare two groups (undergraduate and graduate) at three specific points in time, and grouped bars make that comparison easy because the audience can visually compare each bar height for each year, and across years at a glance. The years (2021, 2023, 2025) are not consecutive, so the bars show each year as its own snapshot. The chart shows that undergraduate enrolment is higher than graduate in every year, and that both decline in 2025. A double line graph would also work to better emphasize the trend.

Comparing **undergraduate** student enrolment by place of residence, grouped into Nova Scotia, Rest of Atlantic Canada (NB, PE, NL), Other Canadian, and International, for the years 2020, 2022, and 2024. See Table 2 and Figure 2

Table 2: FCS undergraduate enrollment by place of residence

Faculty of Computer Science Undergraduates Enrollment figures by Place of Permanent Residence			
Academic Year	Place of Residence		
	Atlantic Canada	**Other Canadian**	**International**
2020/2021	524	659	803
2022/2023	574	744	962
2024/2025	643	835	696

Sources:
Dalhousie University, Registrar's Office. (2020). 2020 enrollment by place of permanent residence [PDF Report]. 2020 Report
Dalhousie University, Registrar's Office. (2020). 2020 enrollment by place of permanent residence [PDF Report]. 2022 Report
Dalhousie University, Registrar's Office. (2020). 2020 enrollment by place of permanent residence [PDF Report]. 2024 Report

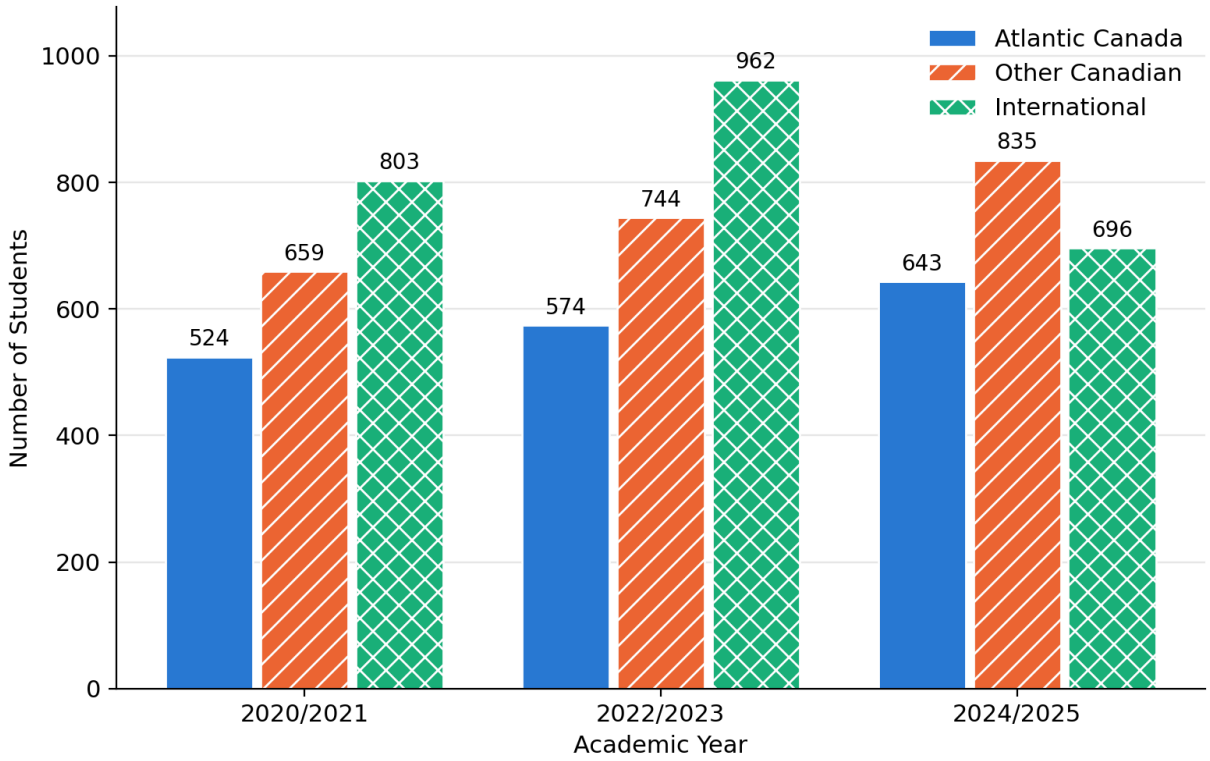


Figure 2: FCS Undergraduate Enrolment by Place of Permanent Residence

Justification for Grouped Bar Chart

The bar chart suits this data because the categories are distinct and the years are non-consecutive (2020/21, 2022/23, 2024/25). Grouping by year lets viewers compare the three residence categories within a year, and the consistent colours serve as a visual cue across years. From the chart it's easy to see Atlantic Canada and Other Canadian enrolment rise steadily, while International peaks in 2022/23 (962) and drops to 696 in 2024/25.

Compare enrolments by majors, first for undergraduates, then for graduates, for the years 2019 - 2025. See Table 3.1 and 3.2. See Figure 3.1 and 3.2

Table 3.1: FCS Ungraduate enrollment by majors, 2019-2025

Faculty of Computer Science

2019-2025 Undergraduate figures by Majors

Majors	Undergraduates						
	2019	**2020**	**2021**	**2022**	**2023**	**2024**	**2025**
Applied Computer Science	243	278	302	357	402	399	363
Computational Biology & Bioinformatics	-	0	0	0	0	0	0
Computer Science	1,046	1,183	1,220	1,349	1,321	1,136	904
Digital Innovation	-	0	0	0	0	0	0
Electronic Commerce	0	0	-	-	-	-	-
Health Informatics	0	0	0	-	-	-	-
Informatics	2	1	-	-	-	-	-
Software Systems	1	-	-	-	-	-	-
Physics	-	-	1	1	1	-	-

Sources:
Dalhousie University, Registrar's Office. (2026). *Registrar's Office: Enrolment reports*. https://www.dal.ca/study/registrars-office.html

Legend

- Major Discontinued
- Major Available
- Major Unavailable

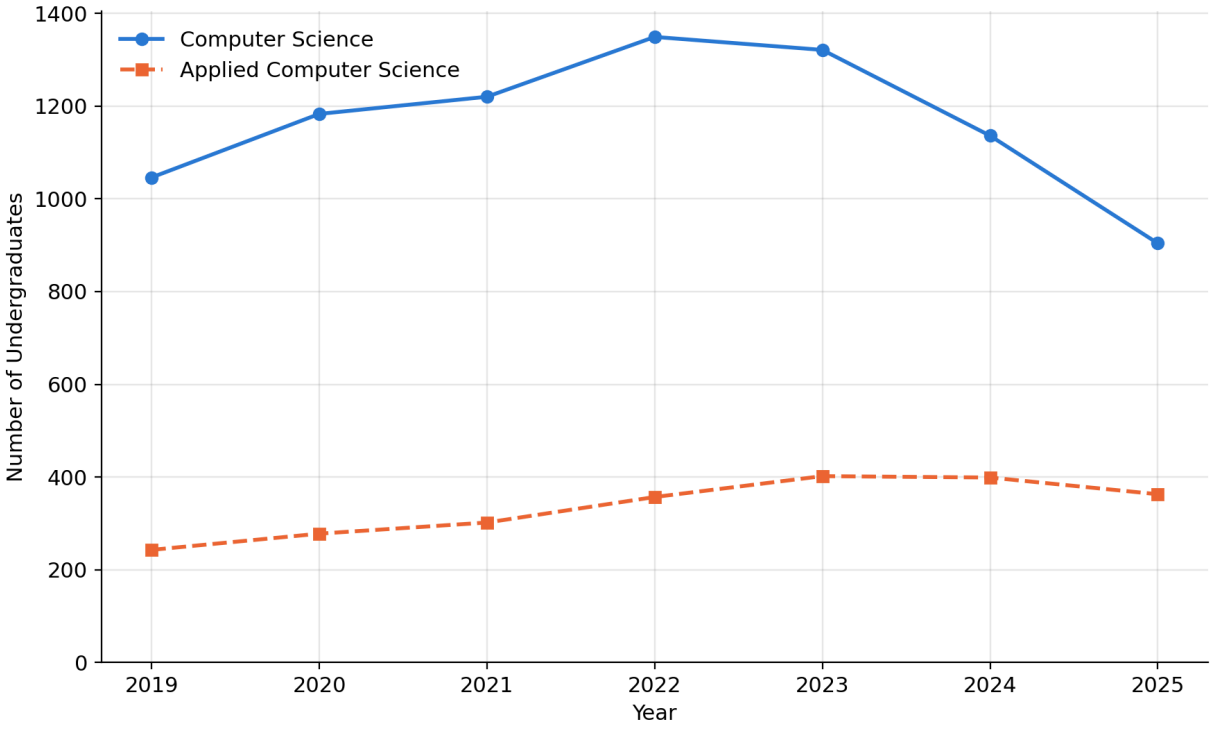


Figure 3.1: FCS Undergraduate Enrolment by Major, 2019-2025

Justification for Multi Line Graph

The data here covers seven consecutive years across several majors, and line graphs show trends, peaks and declines much better in 7 years than bars would. Each line is a separate series(majors). Different line styles, colour, and markers are used, so the lines stay distinguishable.

The chart shows that Computer Science peaks at 1,349 in 2022 and then falls to 904 in 2025, while Applied Computer Science rises to 402 in 2023 and eases to 363. I plotted only these two majors. The others have 0–2 students or are discontinued, so they would show as flat lines along the bottom and add clutter without adding information.

Table 3.2: FCS Graduate enrollment by majors, 2019-2025

Faculty of Computer Science

2019-2025 Graduate Enrollment figures by Majors

Majors	Graduates						
	2019	**2020**	**2021**	**2022**	**2023**	**2024**	**2025**
Applied Computer Science	-	-	-	-	-	-	-
Computational Biology & Bioinformatics	-	1	1	3	4	5	2
Computer Science	297	359	481	580	560	439	282
Digital Innovation	-	27	72	93	100	123	93
Electronic Commerce	38	20	-	-	-	-	-
Health Informatics	39	28	3	-	-	-	-
Informatics	0	1	-	-	-	-	-
Software Systems	0	-	-	-	-	-	-
Physics	-	-	0	0	0	-	-

Sources:
Dalhousie University, Registrar's Office. (2026). *Registrar's Office: Enrolment reports*. https://www.dal.ca/study/registrars-office.html

Legend

- Major Discontinued
- Major Available
- Major Unavailable

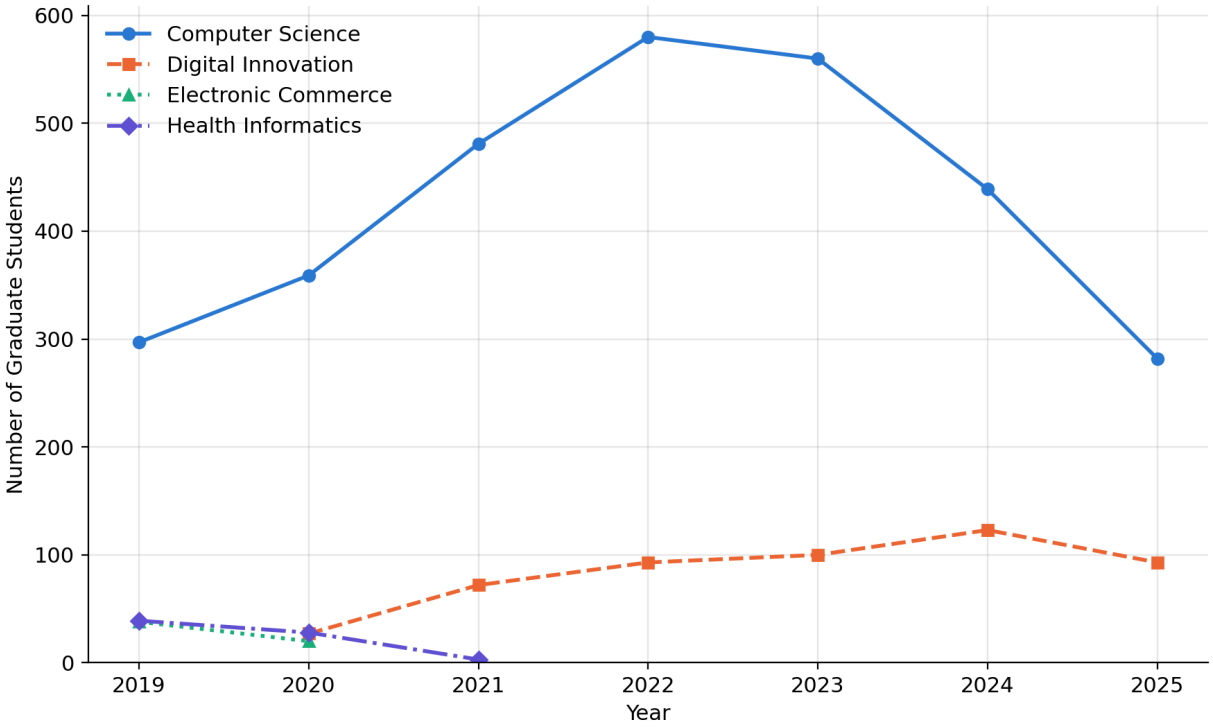


Figure 3.2: FCS Graduate Enrolment by Major, 2019-2025

Justification for Grouped Bar Chart

The reasons are the same as for Figure 3.1